MANAV BAROT

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Summary

Ph.D. candidate and ML engineer specializing in computer vision, LLMs, and multimodal AI. Build scalable PyTorch systems for training, evaluation, and deployment across visual, language, and time-series data.

Research interests: multimodal learning and vision-language reasoning | visual representation learning and generative modeling for image and video understanding | agentic, tool-augmented LLM systems for real-world decision support | 3D human pose and motion understanding.

Education

- **Ph.D. Candidate, Systems Science** | SUNY Binghamton — Aug. 2024 – Dec. 2027 (Expected)
  GPA 3.8/4.0 | Research: vision-language models and grounded LLM systems
- **M.E., Computer Science and Engineering** | Nirma University, Ahmedabad, India — May 2024
  GPA 3.8/4.0 | Thesis: AI-enhanced pose refinement and transition guidance
- **B.E., Computer Science and Engineering** | S.P.B. Patel Engineering College (GTU) — Jun. 2022
  GPA 3.9/4.0 | Thesis: vision-based American Sign Language detection with deep neural networks

Technical Skills

- **Languages:** Python, C, C++, JavaScript
- **Frameworks & Libraries:** PyTorch, PyTorch Lightning, TensorFlow, Hugging Face Transformers, PEFT, NumPy
- **Large-Scale Training & Infrastructure:** Distributed multi-GPU training (DDP, FSDP), DeepSpeed, mixed-precision training, vLLM, Git, Docker, Linux, AWS EC2/S3, GCP

Selected Research Projects

- **PoseVLM: Teaching Vision-Language Models to Reason About Body Geometry**
  - Built a multimodal reasoning system that grounds visual representations in body meshes estimated by SAM 3D Body, enabling joint-angle, symmetry, alignment, and pose-correction reasoning from images
  - Created **YogaAlign**, extending Yoga-82 with image-level 3D body geometry, interpretable biomechanical measurements, and 10,090 instruction-tuning examples spanning pose recognition, comparison, and corrective feedback
  - Introduced a hybrid discrete–continuous decoder for LLaVA-1.5-7B, combining learned geometry-slot tokens for language generation with a regression head for precise numerical prediction
  - Achieved **17.65° mean angular error**, reducing error by **59–62%** over LLaVA-1.5 and Qwen2.5-VL, while improving complete-measurement accuracy from 66.1–68.1% to **86.1%**
- **PoseFix: Visual-Relational Reasoning for Pose Correction**
  - Formulated pose correction as paired-image relational reasoning: infer the directional transformation from an observed posture to a target posture rather than caption either image independently
  - Designed a shared-weight MaxViT dual encoder with stacked bidirectional cross-attention, paired with a transformer decoder that emits actionable corrections
  - Improved ROUGE-L 34.82→44.06, METEOR 21.30→36.80, and CIDEr 6.40→11.39 over published baselines on the FixMyPose benchmark (7,000 image pairs / 11,000 captions)
- **HierarchicalNets: Fine-Grained Classification of Yoga Poses** *Scientific Reports, 2026* **[GitHub]**
  - Investigated hierarchical representation learning for fine-grained pose recognition, where visually similar postures are separated only by relational differences in body configuration
  - Designed four vision-transformer architectures encoding dependencies across a 3-level taxonomy, letting coarse category structure regularize fine-grained discrimination
  - Established new Top-1 results on Yoga-82 at every hierarchy level: 96.79% / 95.64% / 93.07% vs. prior SoTA of 89.81% / 85.10% / 79.35%
- **PhyViT-GAN: Physics-Guided Generative Vision** *IJAMT (Springer), 2026* **[GitHub]**
  - Built PhyViT-GAN, a physics-guided generative vision model that translates simulated component-displacement trajectories into realistic future-state inspection images, bridging numerical physics modeling and image synthesis.
  - Designed a VAE-GAN with a 1D-CNN trajectory encoder, skip-connected image generator, and MobileViT discriminator; achieved an SSIM of 0.8348 and an LPIPS of 0.0917 on held-out data.
- **FIAT: Fourier-Informed Attention Transformer** *In final revision, Applied Energy*
  - Designed an attention module that computes query-key correlation in the frequency domain (FFT, conjugate multiplication, inverse FFT) and aggregates time-delayed shifts, so periodic dependencies are recovered directly rather than learned token by token

– Outperformed statistical (LEAR, GARCH-family) and deep learning (GRU, LSTM) baselines on MAPE and sMAPE across five regional markets, with stable accuracy through high-volatility periods

Research Experience

- **Ph.D. Researcher, R&D** | WISE Research Group (Foxconn Industrial Internet) Dec. 2025 – Present | WI, USA
  - Built an LLM agent combining constrained decoding with retrieval over live multi-sensor telemetry to generate root-cause hypotheses, reducing manual diagnosis time
  - Design and deploy computer vision and anomaly-detection models processing millions of real-time sensor readings for fault detection across a large-scale industrial network
  - Developed spatiotemporal anomaly-detection models over high-throughput, multi-sensor time-series data, improving early-fault detection accuracy at scale
- **Ph.D. Researcher, R&D** | WISE Research Group (Koh Young America) Aug. 2024 – Aug. 2025 | NY
  - Developed 3D vision algorithms for optical inspection (SPI/AOI), focusing on generative modeling, defect representation learning, and sub-pixel spatial localization
  - Proposed **PhyViT-GAN**, a physics-guided generative vision model that learns simulated motion trajectories to predict physically consistent future states, improving sample efficiency under limited data
  - Integrated Siamese and self-supervised defect models into the research pipeline, improving zero-shot generalization to rare defect patterns; validated via Grad-CAM and ablation studies
- **Research Associate** | DoctorOnClick (Singapore) Jun. 2023 – May 2024 | Remote
  - Fine-tuned Mistral-7B and Falcon-7B with RAG for **clinical summarization**, grounding extraction of clinically relevant information in medical records
  - Developed U-Net-based medical-image segmentation and computer-vision pipelines for **physiotherapy gait analysis**, supporting movement assessment and **digital rehabilitation**
- **Computer Vision Engineer** | DoctorOnClick (Singapore) Jan. 2022 – May 2022 | Remote
  - Built and deployed Yogix.ai, a real-time pose-estimation and movement-quality platform supporting 100+ yoga poses at **∼40 FPS (∼25 ms/frame)** on AWS GPU infrastructure, for a Singapore government-backed **health initiative**
  - Owned the ML lifecycle end to end: data pipelines, keypoint-based pose modeling, temporal action recognition, Dockerized API deployment, GPU autoscaling, and inference optimization for real-time feedback
- **Research Assistant** | DoctorOnClick (Singapore) Jul. 2021 – Dec. 2021 | Remote
  - Developed deep learning and computer vision systems for **medical image analysis**, video-based physiological sensing, and human-motion understanding
  - Built applications for camera-based heart-rate estimation via remote photoplethysmography (rPPG), physiotherapy exercise-quality assessment, and longitudinal patient-progress tracking

Academic Service & Awards

- **Session Chair, FAIM 2025**: coordinated peer review, reviewed 8 manuscripts, and chaired a technical session Jun. 2025
- **STEM Education Channel**: produced 60+ instructional videos on Python, computer science, and applied AI 2022–2023
- **Academic Excellence**: achieved a 10.0/10.0 SGPA during the final two semesters of the M.E. program May 2024

Publications

1. **M. Barot**, S. Kojaku. "PoseVLM: Geometry-Conditioned Vision-Language Reasoning for Fine-Grained Pose Understanding." *Manuscript in preparation.*
2. **M. Barot**, P. Thakkar, V. Ukani, M. A. Mehta, K. Kotecha, A. Abraham, D. Kapil, L. Vanderbloemen. "HierarchicalNets for Multi-Level Hierarchical Classification of Yoga Poses." *Scientific Reports*, 2026. DOI: 10.1038/s41598-026-54558-1.
3. **M. Barot**, P. Thakkar. "PoseFix: Natural-Language Instructions for Human Pose Correction." *Manuscript in preparation.*
4. **M. Barot**, J. Kim, D. Won, S. W. Yoon. "PhyViT-GAN: Physics-Guided MobileViT-GAN for Precise Self-Alignment Image Generation." *IJAMT*, 2026.
5. H. Gajjar, **M. Barot**, et al. "Glaucoma Detection Framework Using DeepLabV3+ Aided MobileViT-V2." *Biomedical Signal Processing and Control* (under review).
6. H. Mansy, **M. Barot**, P. Thakkar, S. W. Yoon, Y. Wang. "Fourier-Informed Attention Transformers: A Novel Approach to Electricity Price Forecasting in New Zealand's Market." *Applied Energy* (in final revision).
7. J. Kim, A. Farrag, **M. Barot**, D. Won. "Enhancing Assembly Quality through Parameter Optimization with an Energy Minimization Simulator and Machine Learning." *FAIM 2025.*
8. V. K. Prasad, **M. Barot**. "FloodNetFormer: A Dual-Stream Transformer for SAR-Based Flood Extent Mapping." *WETWAR 2025.*
9. M. Zuhair, P. Bhattacharya, V. K. Prasad, **M. Barot**, M. Modi. "Analysis of Boosting Mechanisms in Cloud-Based Intrusion Detection Systems." *IC3I 2022.*